

Motor Validation Test Report

Project: BLDC Ceiling Fan Motor V2

Date: 03 July 2026

Overall Status: FAIL

Tests Performed:

- Voltage Regulation: PASS
- Thermal Endurance: FAIL (Peak temperature 94°C; limit 85°C)
- Noise Test: PASS
- Efficiency Test: PASS
- Vibration Test: FAIL (Excessive vibration at 1400 RPM)

Observations:

Thermal rise exceeded specification after 60 minutes.

Abnormal vibration observed near rotor assembly.

Possible Causes:

Bearing misalignment, improper rotor balancing, insufficient cooling.

Recommended Actions:

Inspect bearings, rebalance rotor, repeat endurance test.